
**How do regularization techniques and
output noise influence the trajectory
and stability of interaction recovery
in MLPs, as measured against ground
truth?**

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1 Introduction

Neural networks have become a widely adopted model for a wide range of prediction tasks, yet their internal representations remain largely opaque. A central challenge in interpretable machine learning is understanding not only which features influence a model’s output, but how features combine to do so. Feature interactions are cases where the joint effect of two or more features on the output cannot be decomposed into a sum of their individual contributions. These feature interactions are a primary source of this complexity and are of direct scientific interest in domains where understanding the structure of relationships is as important as predictive accuracy (Tsang et al., 2020a; Enouen and Liu, 2023). In practice, correctly identifying which feature combinations drive a prediction is critical in high-stakes settings such as credit scoring, medical diagnosis, and algorithmic hiring, where regulators and practitioners increasingly require explanations that go beyond individual feature importance (Tsang et al., 2020b).

Detecting these interactions reliably is therefore a prerequisite for meaningful interpretability. Neural Interaction Detection (NID), proposed by Tsang et al. (2017), addresses this by reading interaction structure directly from the trained weight matrices of a multilayer perceptron (MLP). Instead of searching through all possible feature subsets, NID uses the observable statistical interactions routed in shared hidden units. NID then uses the strength of the weight connecting features to output as a proxy to measure interaction strength. Because NID operates on the learned weights rather than on model predictions, its effectiveness is directly tied to the structure that training instills in those weights.

This creates a dependency that has received limited attention. The weight structure of a trained network is shaped by every design decision made during training: the optimiser, the regularisation scheme, the level of noise present in the training targets, and the network architecture. If dropout suppresses certain weight paths to prevent overfitting, it may simultaneously suppress the paths that encode interactions. If label noise prevents the model from accurately fitting the target signal, interaction structure may never be strongly encoded in the weights in the first place. While Lengerich et al. (2021) establish that dropout and L_2 regularisation shape interaction learning in distinct ways, and Arpit et al. (2017) show that label noise fundamentally alters the training trajectory, no work has directly measured how these two families of intervention jointly affect the fidelity of interaction recovery against a known ground truth, or how recovery evolves across training epochs rather than at a single convergence point.

This study addresses that gap. Using ten synthetic benchmark functions with known ground truth interaction sets, adopted from Tsang et al. (2017), a factorial experiment is conducted across five output noise levels ($\eta \in \{0, 0.1, 0.2, 0.5, 1.0\}$), two optimisers, two regularisation configurations (dropout and weight decay), and two MLP architectures. NID is applied not only at the best-validation-loss checkpoint but at every saved epoch, producing a full recovery trajectory for each condition. Three dimensions of interaction recovery are examined: the *recovery level*, defined as the AUPRC achieved at the best epoch; the *recovery trajectory*, defined as the sequence of AUPRC values across training epochs; and *stability*, defined as the consistency of recovery across the ten benchmark functions under the same experimental condition, measured by the standard deviation of per-function AUPRC.

The main findings are as follows. Dropout substantially reduces interaction recovery, lowering mean AUPRC by 22.9% relative to an unregularised baseline, while weight decay is considerably more benign (−7.1%). Output noise lowers the attainable recovery plateau without fundamentally changing the early trajectory shape, suggesting that interaction structure is encoded during the early phases of training before label noise fitting begins to compete for representational capacity. Validation loss is a weak proxy for interaction recovery quality ($\rho = -0.22$, $p < 0.001$), implying that standard loss-based model selection does not reliably identify models with strong interaction structure. Smaller architectures match or outperform the deeper model on the majority of benchmark functions, with the largest gaps appearing under adverse conditions.

The code used to generate the data, train the models, and evaluate interaction recovery is publicly available.¹

¹<https://github.com/EdouardDw/MLP-Interaction-Detection>

2 Literature Review

2.1 Feature interactions and why they matter

In the context of neural networks, statistical interactions are formally defined with respect to a set of features I . An interaction exists if the model function $f(x)$ cannot be decomposed into a sum of arbitrary subfunctions that each exclude at least one variable from I (Enouen and Liu, 2023; Tsang et al., 2020a). Mathematically, for a feature set $I = \{x_1, \dots, x_k\}$, a non-additive interaction is present if:

$$f(x) \neq \sum_{i \in I} f_i(x_{I \setminus \{i\}}) \quad (1)$$

This means the entire feature set must be known simultaneously to correctly predict the output, meaning that the features do not act independently (Enouen and Liu, 2023). A simple example is a multiplicative term $(x_1 \cdot x_2)$, which cannot be represented by adding separate functions of x_1 and x_2 alone (Tsang et al., 2020a). It is important to distinguish these non-additive interactions from *pure interactions*, which measure the variance explained by a group of variables that cannot be explained by any strict subset of those variables (Lengerich et al., 2021).

Detecting interactions reliably matters beyond academic interest. In domains such as personalised recommendation and ad targeting, interactions between features often drive the most important predictions, and understanding them is necessary for auditing model behaviour (Tsang et al., 2020a). More broadly, interpretability requirements in applied settings increasingly demand not just which features are important, but how they combine; a model that produces plausible feature importance scores may still misattribute predictions if the underlying interaction structure is incorrectly recovered. Synthetic datasets with known ground truth interactions provide the controlled environment needed to evaluate detection methods rigorously, since researchers can verify whether recovered interactions match the true ones before the method is deployed on real data (Agarwal et al., 2024; Wong et al., 2021).

2.2 Methods for detecting interactions in neural networks

Several families of methods exist for detecting feature interactions in neural networks, and they differ substantially in what they measure and how they measure it. Prediction-based methods such as Friedman’s H-statistic (Friedman and Popescu, 2008) quantify interactions by measuring how much the joint partial dependence of two features deviates from the sum of their individual effects. These methods are model-agnostic but computationally intensive and sensitive to the choice of reference distribution (Friedman and Popescu, 2008).

Attribution-based methods, including Shapley interaction values and integrated gradient decompositions (Lundberg and Lee, 2017), distribute a model’s output back to input pairs and provide theoretically grounded interaction estimates, but typically require many forward passes and can be unstable for higher-order interactions (Enouen and Liu, 2023).

The third family of interaction detection builds interpretability methods directly into the model architecture. Sparse interaction additive networks (Enouen and Liu, 2023) and neural additive models (Agarwal et al., 2021) constrain the model structure to make interactions explicit by design. Since this interpretability is by model design, it cannot be used post hoc on a pre-trained model.

The fourth family is proposed by Tsang et al. (2017) and uses weights to read interaction structure. Neural Interaction Detection (NID) detects interactions directly from the learned parameters of a trained feedforward network. The core insight is that statistical interactions must be rooted through shared hidden units; any interaction present in the network’s function requires at least one hidden unit whose ancestor set at the input layer contains all members of that interaction (Tsang et al., 2017). NID scores candidate interactions by aggregating the product of upstream and downstream weight magnitudes, using the minimum weight in an interaction path as a bottleneck measure. Compared to prediction-based and attribution-based alternatives, NID is fast and requires no additional forward passes, making it practical for trajectory evaluation across many epochs.

Its dependence on the learned weight structure also makes it directly sensitive to any training intervention that changes how weights are distributed. Importantly, L_1 regularization is required to produce the sparse, interpretable weight paths that make this detection feasible (Tsang et al., 2020a), without sparsity, many paths carry non-negligible weight and the interaction signal is diluted.

Because interaction detection is an imbalanced retrieval problem, the number of true interactions is small relative to the full candidate space of all possible feature subsets. Ranking-based evaluation metrics are more informative than accuracy-based ones. With $p = 10$ features, the space of possible interactions

of size two or more contains 1,013 candidates, but the synthetic functions used in Tsang et al. (2017) have only three to five ground truth interactions.

Under this imbalance, a detector that assigns high scores to a few true positives before any false positives should be rewarded, which is precisely what the Area Under the Precision-Recall Curve (AUPRC) captures. AUPRC focuses on the positive class and is insensitive to the large number of true negatives that dominate the candidate space, making it the appropriate primary metric for highly imbalanced retrieval tasks of this kind (Davis and Goadrich, 2006).

2.3 Training dynamics: regularisation and interaction learning

The weight structure that NID reads is not fixed by the architecture alone, it is shaped throughout training by the regularisation scheme applied. The theoretical contrast between dropout and weight decay is important for understanding why the two techniques affect interaction recovery differently.

Dropout randomly removes hidden units during each forward pass, preventing any single pathway from becoming overly specialised (Lengerich et al., 2021). From the perspective of interaction learning, this is consequential: statistical interactions require stable co-activation of multiple input features through a shared hidden unit (Tsang et al., 2017). When dropout stochastically disables units, it disrupts the formation of consistent multi-feature pathways, effectively penalising the network for relying on any particular interaction route.

Lengerich et al. (2021) found that dropout acts as an order-specific regularizer that decays the ability of the model to learn higher-order interactions and forces the model to learn simpler additive patterns over more complex ones.

This forces the model to prefer simple additive patterns over complex interactions. The interaction recovery finds fewer higher-order interactions even if there are genuine higher-order interactions in the data. Weight decay applies an L_2 penalty that shrinks all weights towards zero proportionally (Goodfellow et al., 2016). Weight decay thus reduces all weights without removing pathways or disturbing ranking or interaction strengths.

When too large, L_2 can cause instability by collapsing the model toward constant outputs (Lengerich et al., 2021), but at moderate strength it is substantially less destructive to interaction structure than dropout. This contrast between dropout changing which pathways are active and weight decay only changing their magnitude should entail they have different effects on NID interaction recovery.

2.4 Output noise, memorisation, and recovery trajectories

There are two types of noise discussed in the literature. Firstly, *input noise* refers to perturbations added to the feature vector x . This type of noise is used as a form of regularization and is mathematically equivalent to using an L_2 penalty on weights (Goodfellow et al., 2016).

Output noise, also called label noise, refers instead to perturbations added to the target variable y . This study uses the latter: Gaussian noise proportional to the standard deviation of the clean signal is added to the output of each synthetic function. By definition, output noise does not alter the feature distribution; it degrades only the target signal the model is trying to fit. The Goodfellow equivalence does not apply here, and the two regimes cannot be assumed to produce the same training dynamics or the same effects on learned weight structure.

The dynamics of learning under label noise have been studied in the context of memorization. Arpit et al. (2017) show that neural networks learn noisy and clean examples at different times. The model, in early phases of training, learns patterns that generalize across many batches before fitting idiosyncratic or noisy patterns.

The fact that the model learns noise after patterns that generalize over many examples implies that this information is encoded into the weights earlier during training than noise-driven perturbations.

The trajectory of interaction recovery over epochs is as important as final checkpoint evaluation. Evaluating a model at its best validation loss only indicates how well it recovered interaction (Apicella et al., 2026), but misses the full trajectory indicating if the recovery was gradual, stable, or if early learning was later degraded (Toneva et al., 2019). This motivates using AUPRC as a sequence throughout training epochs rather than a single point, which is a methodological addition to previous research.

2.5 Research gap

Existing work has studied feature interaction detection, regularisation, and label noise largely as separate problems. Three specific gaps remain.

First, Lengerich et al. (2021) establishes that dropout and L_2 regularization affect interaction learning in different ways, but no previous study has measured how these methods’ effects on weight structure affect NID’s ability to recover interactions and how much impact they have on the ability of the model to recover ground truth interactions.

Secondly, previous research generally evaluates interaction recovery at convergence or the best validation loss. Interaction recovery trajectory through the training epochs provides insights into how the model learns that endpoint analysis lacks, such as early plateaus, late degradation, or condition-specific timing. Interaction recovery through the training process has not been studied before.

Thirdly, interaction recovery and fidelity of models with combinations of regularization and label noise have not been explored. Studies address either the generalization effect of regularization or memorization dynamics under noise, but not their influence on interpretability.

This study aims to address these three research gaps by using NID to measure interaction recovery through 100 training epochs, under combinations of regularization techniques, output noise, architecture, and optimizers, evaluated against synthetic data with known ground truth interactions.

3 Methodology

3.1 Synthetic Data

Ten synthetic benchmark functions are used to train the multilayer perceptrons. The synthetic datasets have known ground truth feature interactions defined $\mathcal{G} \subseteq 2^{[10]}$, allowing evaluation of interaction detection methods.

These ten benchmark functions are the same ones used in Tsang et al. (2017) *Detecting Statistical Interactions from Neural Network Weights* paper.

Table 1: Synthetic benchmark functions adapted from Tsang et al. (2017)

Function	Formula	Interaction Sets ($\mathcal{G}$)
f_1	$\pi^{x_1 x_2} \sqrt{2x_3} - \arcsin(x_4) + \log(x_3 + x_5) - \frac{x_9}{x_{10}} \sqrt{\frac{x_7}{x_8}} - x_2 x_7$	$\{1, 2, 3\}, \{3, 5\}, \{7, 8, 9, 10\}, \{2, 7\}$
f_2	$\pi^{x_1 x_2} \sqrt{2 x_3 } - \arcsin(0.5x_4) + \log(x_3 + x_5 + 1) + \frac{x_9}{1+ x_{10} } \sqrt{\frac{ x_7 }{1+ x_8 }} - x_2 x_7$	$\{1, 2, 3\}, \{3, 5\}, \{7, 8, 9, 10\}, \{2, 7\}$
f_3	$e^{ x_1 - x_2 } + x_2 x_3 - (x_3^2)^{ x_4 } + \log(x_4^2 + x_5^2 + x_7^2 + x_8^2) + x_9 + \frac{1}{1+x_{10}^2}$	$\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5, 7, 8\}$
f_4	$e^{ x_1 - x_2 } + x_2 x_3 - (x_3^2)^{ x_4 } + (x_1 x_4)^2 + \log(x_4^2 + x_5^2 + x_7^2 + x_8^2) + x_9 + \frac{1}{1+x_{10}^2}$	$\{1, 2\}, \{2, 3\}, \{3, 4\}, \{1, 4\}, \{4, 5, 7, 8\}$
f_5	$\frac{1}{1+x_1^2+x_2^2+x_3^2} + \sqrt{e^{x_4+x_5}} + x_6 + x_7 + x_8 x_9 x_{10}$	$\{1, 2, 3\}, \{4, 5\}, \{6, 7\}, \{8, 9, 10\}$
f_6	$e^{ x_1 x_2 +1} - e^{ x_3+x_4 +1} + \cos(x_5 + x_6 - x_8) + \sqrt{x_8^2 + x_9^2 + x_{10}^2}$	$\{1, 2\}, \{3, 4\}, \{5, 6, 8\}, \{8, 9, 10\}$
f_7	$(\arctan(x_1) + \arctan(x_2))^2 + \max(x_3 x_4 + x_6, 0) - \frac{1}{1+(x_4 x_5 x_6 x_7 x_8)^2} + \left(\frac{ x_7 }{1+ x_9 }\right)^5 + \sum_{i=1}^{10} x_i$	$\{1, 2\}, \{3, 4, 6\}, \{4, 5, 6, 7, 8\}, \{7, 9\}$
f_8	$x_1 x_2 + 2^{x_3+x_5+x_6} + 2^{x_3+x_4+x_5+x_7} + \sin(x_7 \sin(x_8 + x_9)) + \arccos(0.9x_{10})$	$\{1, 2\}, \{3, 5, 6\}, \{3, 4, 5, 7\}, \{7, 8, 9\}$
f_9	$\tanh(x_1 x_2 + x_3 x_4) \sqrt{ x_5 } + e^{x_5+x_6} + \log((x_6 x_7 x_8)^2 + 1) + x_9 x_{10} + \frac{1}{1+ x_{10} }$	$\{1, 2, 3, 4, 5\}, \{5, 6\}, \{6, 7, 8\}, \{9, 10\}$
f_{10}	$\sinh(x_1 + x_2) + \arccos(\tanh(x_3 + x_5 + x_7)) + \cos(x_4 + x_5) + \frac{1}{\cos(x_7 x_9)}$	$\{1, 2\}, \{3, 5, 7\}, \{4, 5\}, \{7, 9\}$

As shown in Table 1, each function maps a ten-dimensional input vector $\mathbf{x} \in \mathbb{R}^{10}$ to a scalar output y , with a known set of ground-truth interaction sets. The functions are designed to stress-test interaction detection across a range of structural conditions: interaction order varies from pairwise (e.g. $\{1, 2\}$) to higher-order (e.g. $\{4, 5, 6, 7, 8\}$ in f_7), some features participate in multiple distinct interactions (e.g. x_5 in f_9), and several functions contain irrelevant features that do not appear in any interaction. This diversity ensures that results are not specific to a single interaction structure.

The input distribution differs slightly across functions. For f_1 , features $x_1, x_2, x_3, x_6, x_7, x_9$ are drawn from Uniform(0, 1) and the remaining features from Uniform(0.6, 1) to ensure numerical stability of the formula. For all other functions, all ten features are drawn independently from Uniform(−1, 1).

To simulate realistic conditions, Gaussian noise is added to the output proportionally to the standard deviation of the clean signal:

$$\tilde{y} = y + \epsilon, \quad \epsilon \sim \mathcal{N}(0, (\eta \cdot \sigma_y)^2) \quad (2)$$

where σ_y denotes the standard deviation of the noise-free output and η controls the noise level. This formulation ensures that the noise level is interpretable as a fraction of signal variance across all functions, regardless of their differing output scales.

Each dataset consists of $N = 30,000$ samples generated with a fixed seed 42 for reproducibility. The noise level η is varied as an experimental condition and is described further in Section 3.2.

3.2 Experimental Conditions

All experiments are conducted across all ten benchmark functions described in Section 3.1. Five factors are varied: output noise level η , optimizer, dropout rate, weight decay and model size. Table 2 summarises all conditions.

Table 2: Experimental factors and their levels. All combinations are tested, yielding 40 conditions in total, each applied to all ten benchmark functions.

Factor	Levels
Noise η	0, 0.1, 0.2, 0.5, 1.0
Optimizer	Adam, SGD
Dropout	None, 0.2
Weight Decay	Disabled, Enabled ($\lambda = 10^{-4}$)
Model Size	Small MLP, Big MLP

3.3 Interaction detection

Interaction detection is done using the Neural interaction detection (NID) framework developed by Tsang et al. (2017). NID detects statistical interactions by directly interpreting the learned weight matrices of a trained feedforward neural network, avoiding an exponential search over all possible interaction candidates.

3.3.1 Theoretical Basis

The core insight of NID is that statistical interactions between features must be routed through common hidden units. Formally, for any interaction $I \subseteq [p]$ present in the network’s learned function, there exists a hidden unit whose set of ancestors at the input layer contains I (Tsang et al., 2017). This follows from the observation that non-overlapping paths in a network are aggregated via weighted summation at the output, which cannot create interactions between features. Interactions can therefore only arise at hidden units where multiple features converge through strongly weighted connections.

3.3.2 Interaction Strength

For a hidden unit i in the first hidden layer, the interaction strength $\omega_i(\mathcal{I})$ of a candidate interaction $\mathcal{I} \subseteq [p]$ is defined as:

$$\omega_i(\mathcal{I}) = z_i^{(1)} \cdot \mu\left(\left|\mathbf{W}_{i,\mathcal{I}}^{(1)}\right|\right) \quad (3)$$

where $\mathbf{W}_{i,\mathcal{I}}^{(1)}$ are the input-to-first-layer weights connecting features $\mathcal{I}$ to hidden unit i , $\mu(\cdot)$ is an averaging function over those weights, and $z_i^{(1)}$ captures the influence of hidden unit i on the final output via the aggregated weights of all layers above:

$$z^{(1)} = |\mathbf{w}^y|^\top \left| \mathbf{W}^{(L)} \right| \dots \left| \mathbf{W}^{(2)} \right| \quad (4)$$

This quantity upper-bounds the gradient magnitude of the hidden unit with respect to the output, providing a Lipschitz-based approximation of the unit’s importance (Tsang et al., 2017). The averaging function is set to $\mu(\cdot) = \min(\cdot)$, which Tsang et al. (2017) show empirically and analytically to best capture interaction strength: an interaction can only be as strong as its weakest participating weight.

3.3.3 Ranking Algorithm

Interaction strengths are aggregated across all hidden units in the first layer using a greedy ranking algorithm. For each hidden unit, the top- j weighted input features are selected for every order $j = 2, \dots, p$, and their interaction strength is accumulated into a global dictionary. The final interaction strength of a

candidate $\mathcal{I}$ is the sum of $\omega_i(\mathcal{I})$ across all hidden units that proposed it. This produces a ranked list of interaction candidates $\{\mathcal{I}_i\}_{i=1}^m$ ordered by descending strength, from which the detected set $\mathcal{D}$ is obtained by applying a cutoff.

3.3.4 Cutoff

The cutoff on the interaction ranking is determined by constructing a Generalized Additive Model (GAM) that incrementally adds the top-ranked interactions. Interactions are added until validation performance plateaus, at which point the included interactions form the final detected set $\mathcal{D} \subseteq 2^{[10]}$.

3.4 Model Architecture and Training

All models are multilayer perceptrons (MLPs) trained on each of the ten benchmark functions described in Section 3.1. Each MLP uses ReLU activations and is trained to minimize mean squared error (MSE) against the noisy target $\tilde{y}$. No main effect networks are used, meaning the full predictive burden is placed on the interaction network. This corresponds to the standard MLP configuration of Tsang et al. (2017).

Two architectural configurations are evaluated to assess the sensitivity of interaction recovery to network capacity:

- **Small:** two hidden layers of 64 units each
- **Big:** four hidden layers with sizes 140, 100, 60, and 20 units, matching the architecture used in Tsang et al. (2017)

All other training hyperparameters are held constant across both configurations and are summarised in Table 3.

Table 3: Fixed training hyperparameters used across all experiments.

Hyperparameter	Value
Number of samples	30,000
Training / validation split	80% / 20%
Batch size	64
Learning rate	0.01
Number of epochs	100
L_1 regularization constant	10^{-4}
L_2 regularization constant	10^{-4} (when enabled)
Dropout rate	0.20 (when enabled)
Random seed	42

The optimizer is either Adam or SGD depending on the experimental condition, as specified in Table 2. When weight decay is enabled, L_2 regularization is applied globally to all model parameters through the optimizer. L_1 regularization is applied exclusively to the weight matrices of the interaction network, accumulating the sum of absolute weights as a penalty added to the MSE loss prior to backpropagation. This selective application of L_1 encourages sparse weight paths in the interaction network, which is a prerequisite for interpretable interaction detection (Tsang et al., 2017).

Model weights are saved at every epoch, allowing interaction detection to be applied at any point along the training trajectory. For the main analysis, NID is applied to the checkpoint achieving the lowest validation loss. All experiments are conducted with a fixed random seed of 42, applied to both data generation and model initialisation, ensuring full reproducibility.

All 40 experimental conditions described in Section 3.2 are run independently for each architecture, yielding 800 experimental runs in total (40 conditions $\times$ 10 functions $\times$ 2 architectures). The two architectures are not jointly optimized or compared during training. Thus each is trained from scratch under its own conditions. Results are compared post-hoc to assess whether interaction recovery is sensitive to network capacity, as reported in Section 4.

3.5 Evaluation Metrics

To evaluate the quality of interaction detection, interaction recovery is framed as a retrieval problem. For each experimental condition, NID produces a set of detected interactions $\mathcal{D} \subseteq 2^{[10]}$, each accompanied

by an interaction strength score $\omega(I)$. These detections are compared against the known ground truth $\mathcal{G} \subseteq 2^{[10]}$ to construct a binary classification problem: a detection is a true positive if it exactly matches a ground truth interaction, and a false positive otherwise. Superset detections, where NID proposes an interaction that contains a ground truth interaction as a strict subset, are treated as false positives, as they implicate spurious features beyond those in the true interaction.

The evaluation set is the full interaction space $2^{[10]}$, comprising all possible feature subsets of size $|I| \geq 2$. With $p = 10$ features, this yields 1,013 candidate interactions in total. For each candidate interaction $I \subseteq 2^{[10]}$ with $|I| \geq 2$, a binary label and a score are assigned as follows:

$$y_I = \begin{cases} 1 & \text{if } I \in \mathcal{G} \\ 0 & \text{if } I \notin \mathcal{G} \end{cases} \quad (5)$$

and score:

$$s_I = \begin{cases} \omega(I) & \text{if } I \in \mathcal{D} \\ 0 & \text{if } I \notin \mathcal{D} \end{cases} \quad (6)$$

This scoring scheme penalizes both missed detections and false positives. A ground truth interaction $I \in \mathcal{G} \setminus \mathcal{D}$ that NID fails to propose receives a score of zero, placing it at a disadvantage relative to any false positive with a non-zero strength. A non-interaction $I \notin \mathcal{G}$ that NID does not detect receives score zero with label zero, contributing to the true negative class. The four resulting cases are summarized in Table 4.

Table 4: Confusion matrix for evaluation over the full interaction space $2^{[10]}$.

		Actual	
		$I \in \mathcal{G}$	$I \notin \mathcal{G}$
Predicted	$I \in \mathcal{D}$	$\mathcal{G} \cap \mathcal{D}$ (TP)	$\mathcal{D} \setminus \mathcal{G}$ (FP)
	$I \notin \mathcal{D}$	$\mathcal{G} \setminus \mathcal{D}$ (FN)	$2^{[10]} \setminus (\mathcal{G} \cup \mathcal{D})$ (TN)

The primary evaluation metric is the Area Under the Precision-Recall Curve (AUPRC), computed over the full interaction space. AUPRC is appropriate here because the number of true interactions is very small relative to the full candidate space of 1,013 subsets, making the task highly imbalanced. It directly evaluates whether ground truth interactions receive high scores before false positives accumulate, which is precisely the property of interest when assessing interaction recovery fidelity. An alternative ranking-based metric, AUROC evaluated over the restricted set $\mathcal{G} \cup \mathcal{D}$, was also computed for all conditions. The two metrics are strongly correlated (Spearman $\rho = 0.968$), confirming that the qualitative findings are robust to the choice of evaluation formulation.

AUPRC is computed by sweeping a threshold over the ranked score distribution and recording precision and recall at each point. For a threshold τ , the predicted positive set is $\hat{\mathcal{D}}_\tau = \{I : s_I \geq \tau\}$, giving:

$$\text{Precision}(\tau) = \frac{|\hat{\mathcal{D}}_\tau \cap \mathcal{G}|}{|\hat{\mathcal{D}}_\tau|} \quad \text{Recall}(\tau) = \frac{|\hat{\mathcal{D}}_\tau \cap \mathcal{G}|}{|\mathcal{G}|} \quad (7)$$

The curve is formed by varying τ from the highest to the lowest observed score, and the area is computed using the trapezoid rule. The curve is anchored at recall = 0, precision = 1 and closed at recall = 1, precision = $|\mathcal{G}|/|2^{[10]}|$, corresponding to the random baseline. Formally:

$$\text{AUPRC} = \int_0^1 \text{Precision}(r) dr \quad (8)$$

The random baseline AUPRC equals the prevalence of positive interactions in the full candidate space:

$$\text{AUPRC}_{\text{random}} = \frac{|\mathcal{G}|}{|2^{[10]}|} \approx \frac{4.11}{1013} \approx 0.004 \quad (9)$$

where $|\mathcal{G}| = 4.11$ is the mean number of ground truth interactions per function across the ten benchmark functions. An AUPRC of 1.0 indicates that every ground truth interaction is retrieved before any false positive, meaning NID perfectly recovers the true interaction structure. Values close to the random

baseline indicate that NID assigns interaction strengths essentially at random with respect to the ground truth. AUPRC is reported as the primary evaluation metric throughout this study, averaged across all ten benchmark functions for each experimental condition.

3.6 Trajectory Evaluation

The primary analysis applies NID to the checkpoint achieving the lowest validation loss, providing a single snapshot of interaction recovery quality at convergence. While this is sufficient for comparing final performance across conditions, it cannot reveal how interaction recovery evolves during training. To study this, NID is applied independently at every saved epoch checkpoint, producing a recovery trajectory for each experimental condition.

Formally, for a given function f and experiment e , a trajectory is a sequence of AUPRC values:

$$\text{AUPRC}_t = \text{AUPRC}(\mathcal{G}, \mathcal{D}_t), \quad t = 1, \dots, T \quad (10)$$

where $\mathcal{D}_t$ denotes the set of interactions detected by NID at epoch t and $\mathcal{G}$ is the fixed ground truth. The ground truth does not change across epochs; only the NID detections vary as the model weights evolve. Each evaluation uses the same scoring procedure described in Section 3.5, applied independently at each checkpoint.

Trajectories are aggregated across functions using the same two-stage procedure used throughout this study: AUPRC values are first averaged within each function across experimental replicates, and the resulting function-level means are then averaged across functions. This produces a mean trajectory accompanied by a ± 1 standard deviation band, where the band reflects variability across functions rather than across epochs.

The peak AUPRC epoch for a given trajectory is defined as the earliest epoch at which the maximum AUPRC value is observed:

$$t^* = \min \left\{ t \mid \text{AUPRC}_t = \max_{t'} \text{AUPRC}_{t'} \right\} \quad (11)$$

This definition is sensitive to epoch-level noise in the AUPRC signal, which contributes to the high standard deviations reported in Table 8. Trajectory data is used in three analyses in Section 4: the interaction recovery grid comparing all noise-by-regularization combinations, the loss–AUPRC divergence analysis contrasting noise = 0 and noise = 1, and the peak epoch analysis examining when recovery plateaus under each condition.

It should be noted that all trajectories are computed from single-seed runs. The observed variability across functions therefore includes genuine structural differences between functions with run-to-run randomness, and conclusions about trajectory shape should be interpreted accordingly.

3.7 Convergence Restriction

Not all experimental runs produced meaningful learning. Under high output noise or strong regularisation, a subset of models failed to reduce their validation loss over the course of training, indicating that the network never extracted usable structure from the data. Including such runs in the analysis would mix up genuine interaction recovery with the behavior of models that simply did not learn, inflating variance and obscuring the effect of the experimental conditions.

To address this, a convergence criterion was applied uniformly across all analyses. For each run, the validation loss at the first recorded epoch was compared to the validation loss at the final epoch (epoch 100). A run was deemed non-convergent if the relative decrease did not meet a minimum threshold:

$$\frac{\ell_1 - \ell_{100}}{\ell_1} < 0.03 \quad (12)$$

where ℓ_1 and ℓ_{100} denote the validation loss at epoch 1 and epoch 100 respectively. Any run satisfying this condition was excluded. The threshold of 3% was chosen to be conservative: it removes runs that are clearly flat while retaining runs that converge slowly or noisily, such as those under dropout regularization.

Figure 1 illustrates the motivation for this criterion. At noise = 0, validation loss decreases sharply and AUPRC rises steadily across training, confirming that the model learns meaningful feature interactions. At noise = 1, validation loss remains near its initial value throughout all 100 epochs and AUPRC shows no systematic improvement, indicating a complete failure to extract signal from the data. The contrast between these two conditions makes clear that retaining non-convergent runs would introduce noise without adding information.

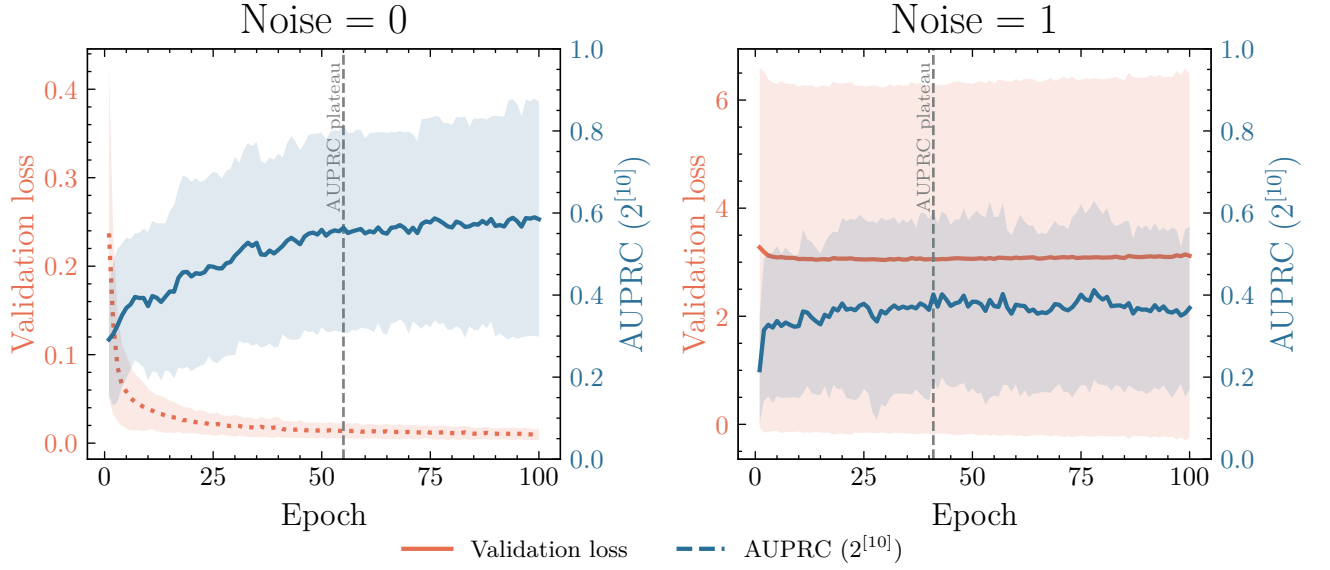


Figure 1: Validation loss and AUPRC (2^{10}) trajectories over 100 epochs for the base condition, without regularisation, on the small model, averaged across all ten synthetic functions. Shaded regions denote ± 1 standard deviation across functions. The left panel shows $\eta = 0$, and the right panel shows $\eta = 1$. The dashed vertical line marks the epoch at which AUPRC reaches 95% of its series maximum.

Figure 2 summarises the removal rates across all noise levels and regularisation conditions. 16% of experiments were removed. The pattern is consistent across both architectures. Weight decay alone is rarely removed, suggesting that L_2 penalisation does not prevent the model from reducing validation loss. Dropout conditions are more frequently removed, particularly at $\eta = 0.2$ and $\eta = 0$, which is consistent with dropout destabilising optimisation in the absence of sufficient noise to regularise against. The combined dropout and weight decay condition shows the highest removal rate at noise = 0, suggesting that the two regularisers in combination over-constrain the network on clean data. At $\eta = 1$, base and dropout conditions are also heavily affected, reflecting the absence of any learnable signal. A run-level breakdown of which specific function and experiment combinations were excluded is provided in Appendix A.

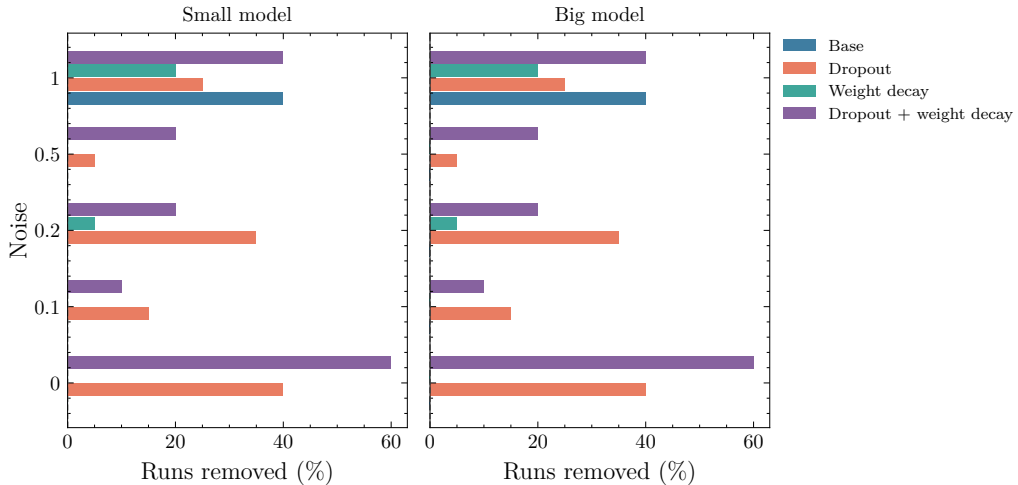


Figure 2: Percentage of runs removed by the convergence filter, broken down by noise level and regularisation condition for the small model (left) and big model (right).

The criterion was applied consistently across all downstream analyses: the best-epoch summary results used for all bar charts and heatmaps, the epoch-level trajectories used for the recovery trajectory plots, and the peak-epoch summaries used for Table 8. The only exception is Figure 1 itself, which intentionally includes all runs to show the filtering.

4 Analysis and Results

4.1 Baseline Interaction Recovery

Across all experimental conditions and both architectures, NID recovers interaction structure well above the random baseline of 0.004 for all ten benchmark functions, with an overall mean AUPRC of 0.462. Table 5 and Figure 3 show, however, that recovery quality varies substantially across functions. f_1 achieves the lowest mean AUPRC (0.121), while f_5 and f_{10} reach the highest (0.692 and 0.727 respectively). This six-fold range confirms that function identity is a strong source of variance in the results and motivates the two-stage aggregation procedure used throughout this section, which averages within functions before averaging across them.

Despite the variation in plateau height, the trajectory shape is consistent across functions. As shown in Figure 3, all functions exhibit a rapid rise in AUPRC during the first 10–20 epochs, followed by a stable plateau that persists through epoch 100. This suggests that the interaction recovery process itself follows a common structural pattern regardless of function complexity, and that the conditions under which this process operates (regularisation, noise, and architecture) primarily affect the ceiling rather than the shape of recovery.

The large standard deviations for f_5 , f_8 , and f_{10} (0.189, 0.278, and 0.233 respectively) indicate that these functions are particularly sensitive to experimental conditions, with some conditions achieving near-perfect recovery (AUPRC = 1.000) while others approach the random baseline. In contrast, f_9 shows the smallest standard deviation (0.078), suggesting its interaction structure is recovered consistently regardless of the experimental condition.

Table 5: Mean AUPRC per synthetic function across all experimental conditions.

Function	Mean AUPRC	SD	Min	Max
f_1	0.121	0.100	0.004	0.526
f_2	0.272	0.089	0.086	0.480
f_3	0.414	0.166	0.253	0.804
f_4	0.442	0.185	0.204	0.858
f_5	0.692	0.189	0.253	1.000
f_6	0.507	0.162	0.253	0.854
f_7	0.380	0.122	0.253	0.688
f_8	0.580	0.278	0.004	1.000
f_9	0.489	0.078	0.004	0.595
f_{10}	0.727	0.233	0.004	1.000

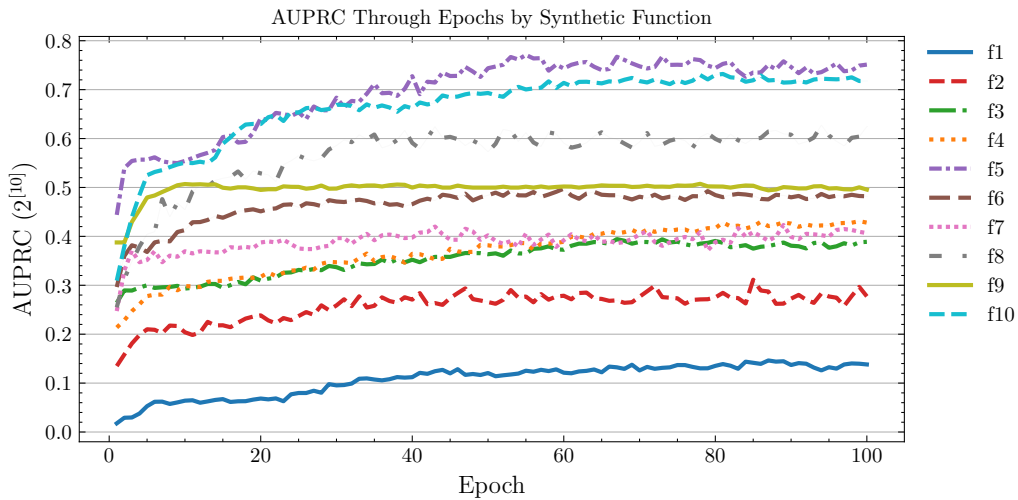


Figure 3: AUPRC trajectories over 100 epochs for each of the ten synthetic benchmark functions, averaged across all experimental conditions.

4.2 Effect of Regularisation on Recovery Trajectory

All regularisation conditions reduce interaction recovery relative to the base condition, but to different degrees. Table 6 shows that weight decay is the least damaging, reducing mean AUPRC by only 7.1% relative to base (0.486 vs 0.524). Dropout and the combined dropout and weight decay condition produce substantially larger reductions of 22.9% and 23.7% respectively, converging to nearly identical mean AUPRC values of 0.404 and 0.400. The near-equivalence of dropout alone and dropout combined with weight decay suggests that dropout is the dominant factor in suppressing recovery, with weight decay adding little additional cost once dropout is already applied.

Figure 4 shows that this ordering happens in the early phase of model training. The gap between base and dropout conditions opens within the first five epochs and remains stable throughout training. All conditions follow the same rapid early rise seen in the baseline analysis, but the dropout conditions plateau at a lower ceiling. This is consistent with the finding of Lengerich et al. (2021) that dropout acts as an order-specific regulariser, decaying the effective learning rate of higher-order interactions and preventing the model from fully exploiting the interaction structure present in the data.

Figure 5 shows that the small model outperforms the big model for 3 out of 4 regularization conditions. The small model consistently outperforms the big model under base and weight decay conditions, while the two architectures converge under dropout, suggesting that the capacity advantage of the big model is negated when dropout prevents effective interaction learning. This architecture regularisation interaction is discussed further in Section 4.9.

Table 6: Mean AUPRC by regularisation condition, averaged across all functions, noise levels, and architectures.

Regularisation	Mean AUPRC	SD	Drop vs Base (%)
Base	0.524	0.219	–
Dropout	0.404	0.171	–22.9%
Weight decay	0.486	0.196	–7.1%
Dropout + weight decay	0.400	0.171	–23.7%

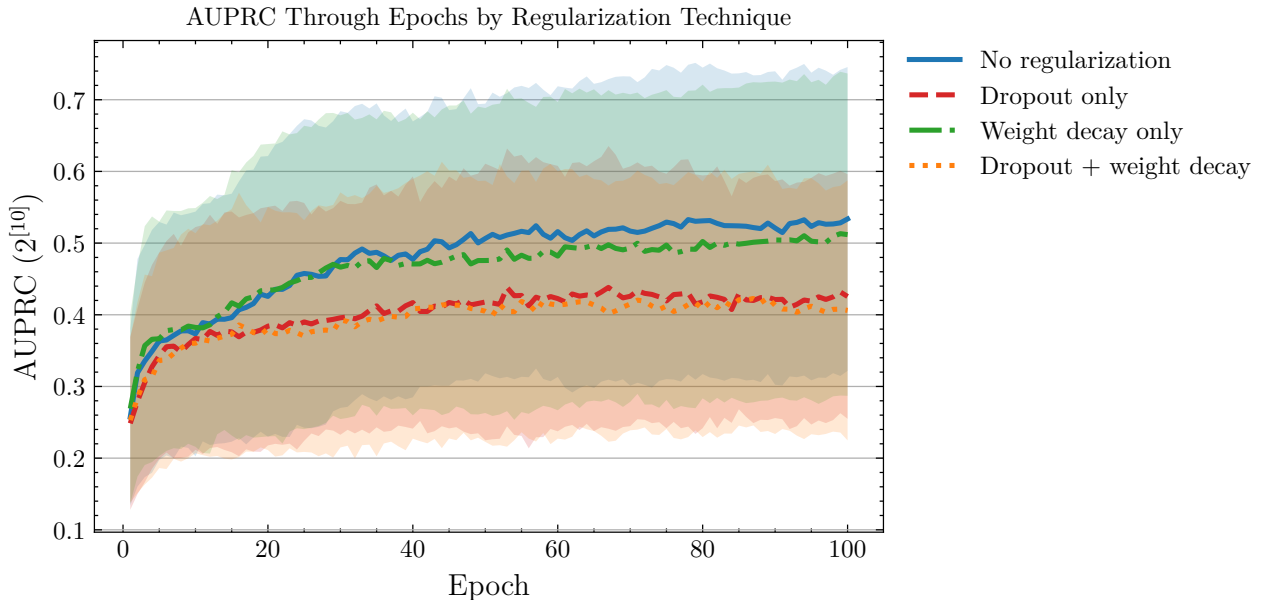


Figure 4: AUPRC trajectories over 100 epochs by regularisation condition, averaged across all functions and noise levels. Shaded regions denote ± 1 standard deviation across functions.

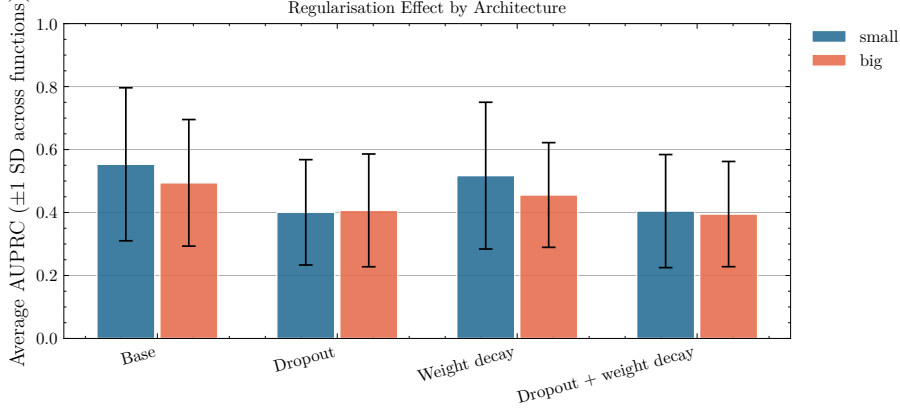


Figure 5: Mean AUPRC at best epoch by regularisation condition and architecture, averaged across all functions and noise levels. Error bars denote ± 1 standard deviation across functions.

4.3 Effect of Regularisation Strength on Interaction Recovery

The preceding analysis fixed dropout at $p = 0.2$ and weight decay at $\lambda = 10^{-4}$. This leaves open whether the 22.9% and 7.1% reductions reported in Table 6 are particular to those two settings or representative of a broader proportional relationship. To probe this, a targeted supplementary sweep was conducted on a representative subset of three benchmark functions (f_1 , f_7 , f_8 , spanning the low-, mid-, and high-recovery range), holding all other factors at their cleanest configuration (small architecture, Adam, $\eta = 0$). Dropout was swept over $p \in \{0, 0.1, 0.2, 0.3, 0.5\}$ and weight decay over $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$, with each condition expressed as the percentage change in best-epoch AUPRC relative to that function’s own zero-strength anchor before averaging across functions.

Figure 6 shows the two sweeps. The clearest result is asymmetric. Strong dropout ($p = 0.5$) reduces recovery by roughly 40% relative to no dropout, extending the $p = 0.2$ finding to higher rates, but the pattern across intermediate values is noisy, and the per-function spread is large. Weight decay produces no reliable degradation at any tested strength: mean recovery stays within roughly $\pm 20\%$ of the anchor across four orders of magnitude of λ , reinforcing the main result that L_2 penalization is benign for interaction recovery. The cross-function standard deviation widens substantially at intermediate dropout rates and at the strongest weight-decay setting, indicating that individual functions respond heterogeneously to regularization strength rather than following a common curve. At three functions and a single seed, these sweeps characterize the general empirical shape of the relationship: clear suppression under strong dropout and tolerance of weight decay. Full per-condition values are reported in Appendix C.

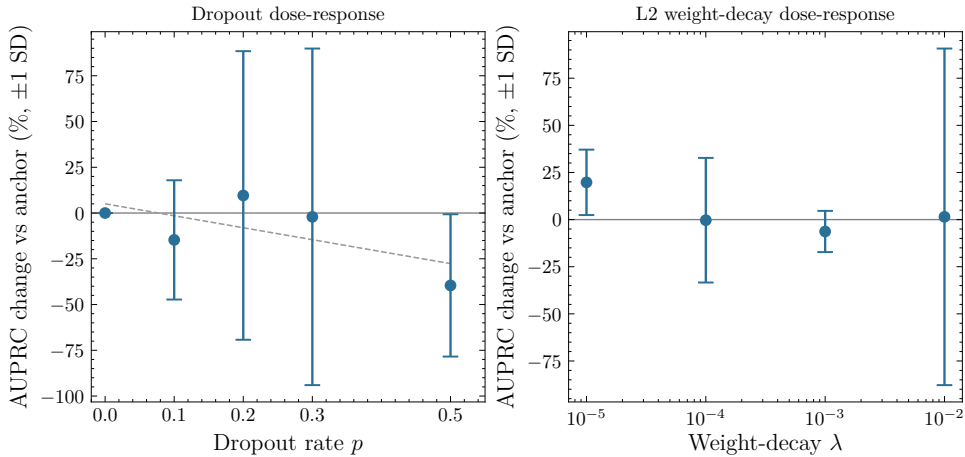


Figure 6: Dose-response of interaction recovery to regularization strength, on the probe subset (f_1 , f_7 , f_8 ; small architecture, Adam, $\eta = 0$). *Left*: percentage change in best-epoch AUPRC versus dropout rate p , relative to $p = 0$. *Right*: percentage change versus weight-decay λ (log axis), relative to $\lambda = 0$. Points are means across the three functions; error bars denote ± 1 standard deviation across functions. Single-seed runs.

4.4 Effect of Noise on Recovery

Label noise reduces interaction recovery across all noise levels, as shown in Table 7. The degradation is gradual at low noise levels. Pooled AUPRC drops by only 12.8% and 13.2% at $\eta = 0.1$ and $\eta = 0.2$ respectively, but accelerates at higher levels, reaching a 36.2% reduction at $\eta = 1$ relative to the noise-free condition (0.360 vs 0.564). The step between $\eta = 0.2$ and $\eta = 0.5$ (−13.2% to −24.7%) is notably larger than the step between $\eta = 0$ and $\eta = 0.2$, suggesting moderate label noise is tolerated but high noise impairs the model’s ability to encode interaction structure in its weights.

Figure 7 suggests that output noise primarily affects the plateau height of the recovery trajectory rather than its shape. All noise levels follow the same rapid early rise in the first 10–20 epochs, after which trajectories diverge to different stable plateaus. This pattern is consistent with the view that the interaction recovery process operates on its own timescale in early training, and that output noise acts mainly as a ceiling constraint on the final recovery level rather than fundamentally disrupting the early recovery dynamics. This is consistent with Arpit et al. (2017), who show that noise fitting occurs as a later phase in training after real patterns have already been learned.

Figure 8 shows an asymmetry between the two architectures that becomes more pronounced at high noise. At $\eta = 0$, the big model marginally outperforms the small model (0.579 vs 0.549). However, at $\eta = 1$ this advantage reverses: the small model achieves 0.395 against the big model’s 0.325. This is consistent with the larger model being more susceptible to label noise, potentially because its greater capacity allows it to fit noisy targets more readily, leaving less representational structure available for genuine interaction recovery.

Table 7: Mean AUPRC by noise level and architecture.

Noise	Small	Big	Pooled	Drop vs $\eta = 0$ (%)
0	0.549	0.579	0.564	–
0.1	0.505	0.478	0.492	−12.8%
0.2	0.503	0.475	0.489	−13.2%
0.5	0.457	0.393	0.425	−24.7%
1	0.395	0.325	0.360	−36.2%

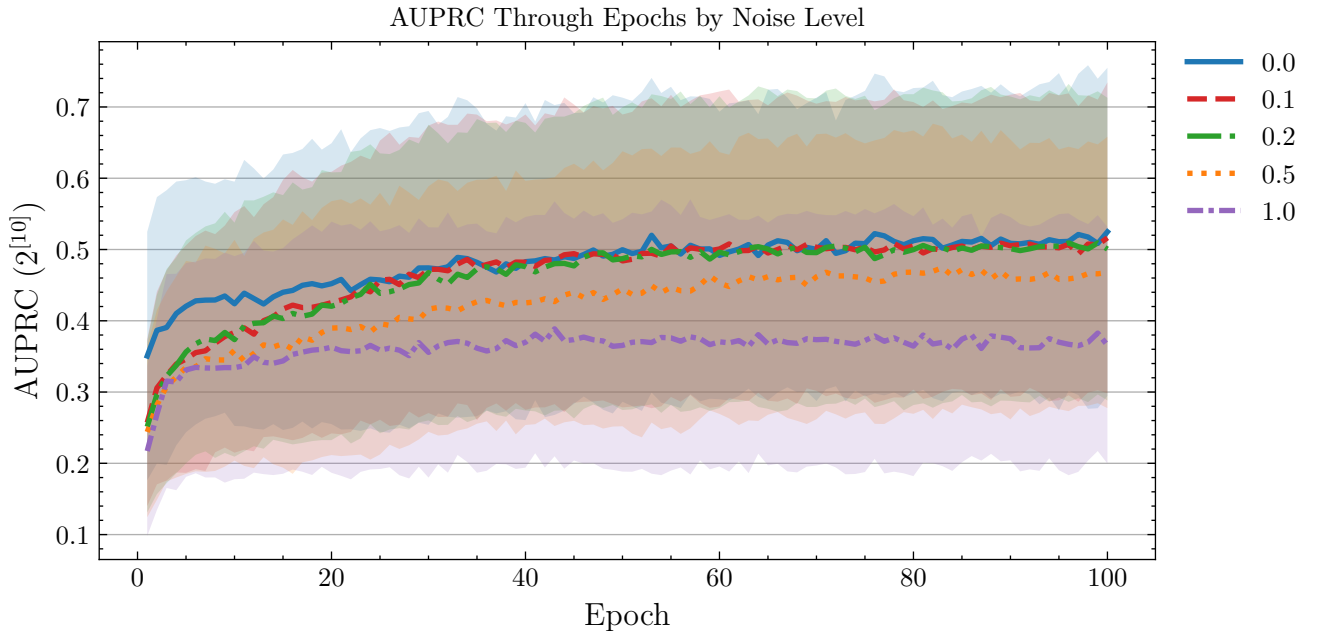


Figure 7: AUPRC trajectories over 100 epochs by noise level, averaged across all functions and experimental conditions. Shaded regions denote ± 1 standard deviation across functions.

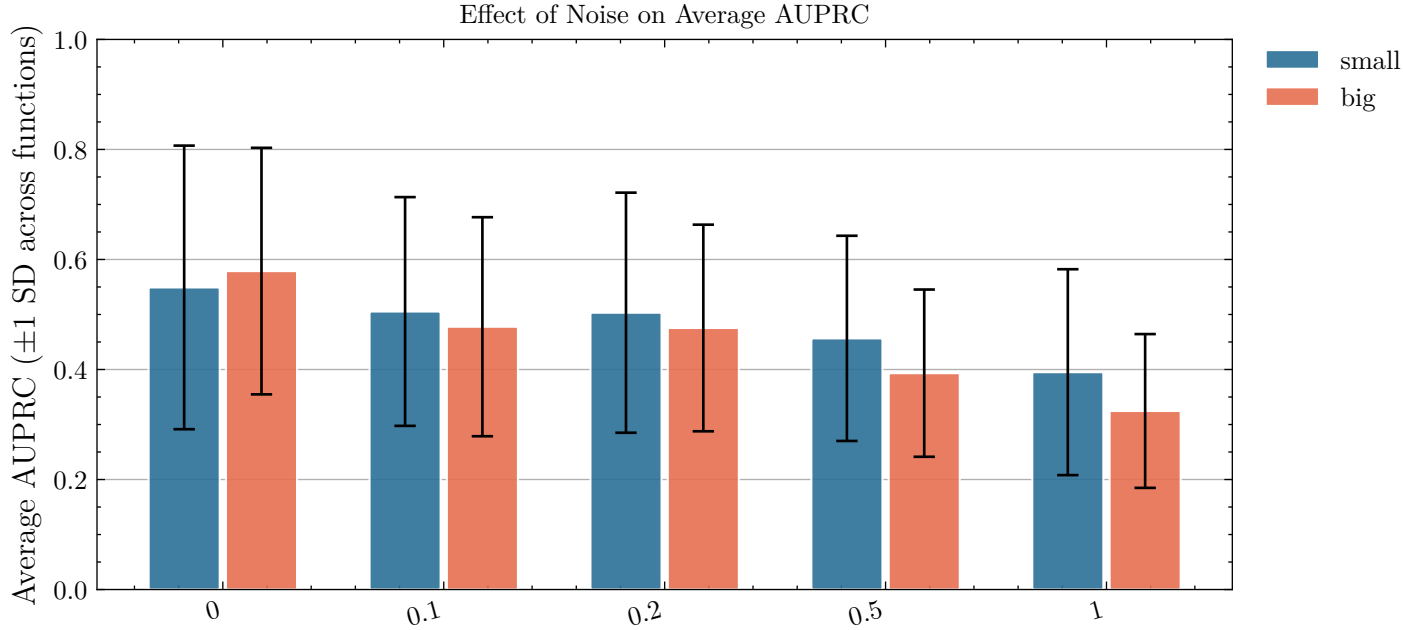


Figure 8: Mean AUPRC at best epoch by noise level and architecture, averaged across all functions and regularisation conditions. Error bars denote ± 1 standard deviation across functions.

4.5 Joint Effect of Noise and Regularization

Figure 9 presents the joint effect of noise level and regularisation condition on mean AUPRC, averaged across all functions and both architectures. The heatmap reveals that the two factors interact somewhat additively: each independently degrades recovery, and the worst cell combines both in their most damaging configurations.

The highest mean AUPRC is achieved by the base condition at zero noise (0.59), and the lowest by weight decay at noise $\eta = 1$ (0.32). Noise and regularisation impose comparable worst-case penalties: the largest noise-induced drop (weight decay, $0.58 \rightarrow 0.32$, a fall of 0.26 from $\eta = 0$ to $\eta = 1$) is essentially tied with the largest regularisation gap at a fixed noise level (base versus dropout + weight decay at $\eta = 0$, $0.59 \rightarrow 0.34$, a gap of 0.25). The effect of noise is, however, highly condition-dependent: it costs 0.20 points under the base condition and 0.26 under weight decay, but leaves the dropout conditions almost unchanged (a drop of only 0.07 for dropout, and a marginal *increase* of 0.02 for dropout + weight decay). Weight decay is thus the most noise-sensitive regularisation condition, whereas dropout is the most noise-robust.

Unlike dropout, weight decay tracks the base condition closely at low to moderate noise, matching it at $\eta \in \{0, 0.1, 0.2\}$ and slightly exceeding it at $\eta = 0.5$ (0.47 vs 0.45). At $\eta = 1$ this advantage disappears and weight decay falls to 0.32, below base, consistent with the absence of any learnable signal at the highest noise level. The broader contrast remains: L_2 penalisation stays competitive with the unregularised baseline across most of the noise range, whereas dropout depresses recovery throughout, possibly because weight decay constrains weight magnitude without disrupting the relative ordering of interaction path strengths that NID reads.

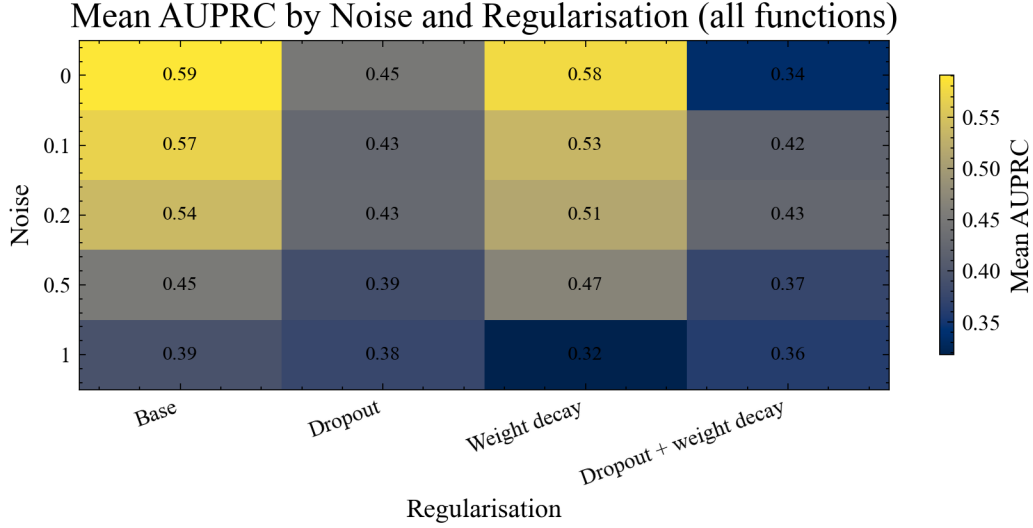


Figure 9: Mean AUPRC by noise level and regularisation condition, averaged across all functions and both architectures.

4.6 Recovery Trajectories Across Conditions

Figure 10 contrasts the four most informative trajectory conditions: base versus dropout plus weight decay, crossed with noise = 0 and noise = 1. Together these four panels illustrate how regularisation and noise shape recovery dynamics rather than simply rescaling the final AUPRC.

At noise = 0 under the base condition (top left), recovery is progressive and sustained. The mean trajectory rises steadily across all 100 epochs, with individual function trajectories spreading over a wide range but consistently improving. This confirms that clean data provides sufficient signal for the network to encode interaction structure throughout training. At noise = 0 under dropout plus weight decay (top right), the trajectory is markedly different in character: the mean is lower and flat from early epochs, with several individual functions plateauing near 0.50 for the entire duration of training. Regularisation in this condition appears to prevent the progressive refinement seen in the base case, consistent with Lengerich et al. (2021)’s observation that dropout decays the effective learning rate of higher-order interactions.

At noise = 1 (bottom row), both conditions converge to similarly low and flat trajectories, though the dynamics differ. The base condition (bottom left) stabilises near 0.35 from the earliest epochs, showing no meaningful improvement across training. The dropout plus weight decay condition (bottom right) shows a more pronounced early rise before plateauing at a comparable level near 0.38. This suggests that regularisation can temporarily facilitate early interaction recovery even under heavy noise, perhaps by preventing premature overfitting to noise patterns, but that this advantage does not translate into sustained improvement. The full trajectory grid across all noise levels and regularisation conditions is provided in Appendix B.

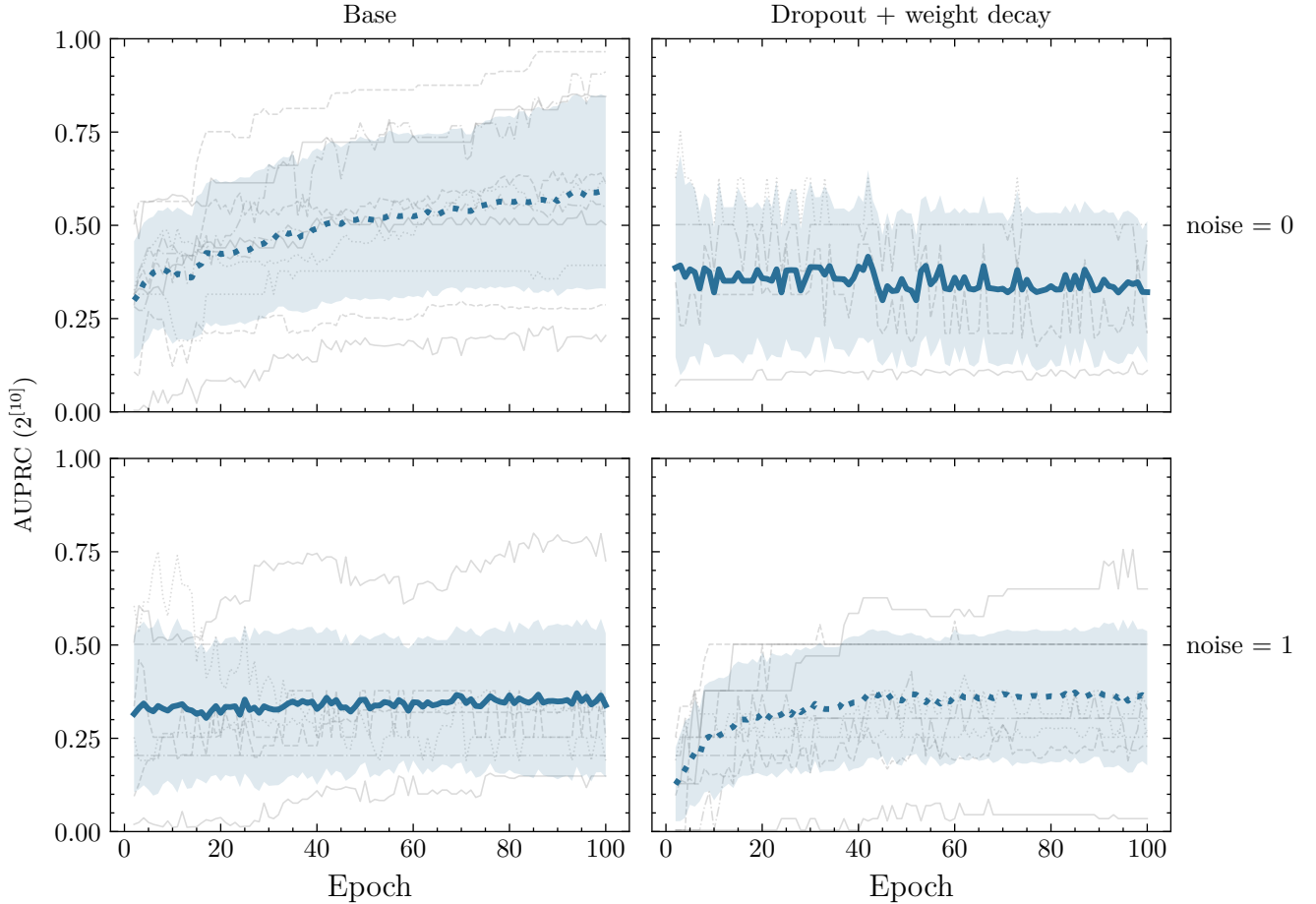


Figure 10: AUPRC recovery trajectories for the base condition and dropout + weight decay condition at noise = 0 (top) and noise = 1 (bottom). Grey lines show individual function trajectories; the bold line shows the mean across functions with ± 1 standard deviation shaded. The full grid across all noise levels and regularisation conditions is provided in Appendix B.

4.7 Timing of Peak Recovery

Table 8 reports the mean epoch at which peak AUPRC is reached for each combination of noise level and regularisation condition. Two patterns emerge clearly from the data.

First, the base condition reaches its peak latest at low noise (40.0 ± 17.4 at $\eta = 0$), while dropout plus weight decay peaks earliest (8.0 ± 10.7 at $\eta = 0$). This ordering is consistent with the trajectory shapes in Figure 10: the base condition continues improving gradually throughout training, whereas heavy regularisation forces an early plateau. Dropout alone shows an intermediate pattern, peaking around epoch 16 at $\eta = 0$ and stabilising near epochs 34–42 for moderate noise levels.

Second, peak timing appears to interact with the regularisation condition, though the large cross-function standard deviations in Table 8 mean these patterns should be read as suggestive rather than firmly established. For the base and weight decay conditions, peak timing is relatively stable from $\eta = 0.1$ to $\eta = 0.5$, clustering in the range of epochs 38 to 46. At $\eta = 1$, both peak earlier (22.4 and 25.9), consistent with the absence of a learnable signal preventing any late-epoch improvement. The dropout conditions do not show this early-peak shift at $\eta = 1$, if anything dropout peaks later.

The large standard deviations throughout the table reflect genuine variability in peak timing across functions rather than measurement noise in the AUPRC signal. Functions with complex interaction structures or higher sensitivity to experimental conditions peak at very different epochs from simpler functions.

Noise level	Base	Dropout	Weight decay	Dropout + weight decay
0	40.0 $\pm$ 17.4	16.1 $\pm$ 23.2	29.6 $\pm$ 30.9	8.0 $\pm$ 10.7
0.1	43.6 $\pm$ 15.2	35.1 $\pm$ 14.7	45.9 $\pm$ 13.3	28.5 $\pm$ 20.1
0.2	38.3 $\pm$ 13.5	42.3 $\pm$ 20.0	42.6 $\pm$ 15.1	37.2 $\pm$ 18.4
0.5	39.0 $\pm$ 18.6	34.4 $\pm$ 21.7	38.8 $\pm$ 25.3	38.6 $\pm$ 21.2
1	22.4 $\pm$ 21.1	38.5 $\pm$ 22.3	25.9 $\pm$ 18.4	36.3 $\pm$ 28.2

Table 8: Mean epoch of peak AUPRC by noise level and regularisation condition, averaged across all functions. Values denote mean $\pm$ standard deviation across functions.

4.8 Optimizer Effect

Adam consistently outperforms SGD on interaction recovery across all conditions. Table 9 shows a mean AUPRC of 0.513 for Adam versus 0.429 for SGD, a gap of 0.084 points. Figure 11 reveals that this difference is not only a final-performance effect but a trajectory effect: Adam reaches a high plateau within the first five epochs and maintains it stably throughout training, while SGD rises more slowly and does not converge to a comparable level even by epoch 100. The shaded bands show that both optimisers exhibit similar cross-function variance, so the gap is not driven by SGD failing on a small number of outlier functions.

The slower convergence of SGD is consistent with its well-known sensitivity to learning rate scheduling (Schaul et al., 2013). With a fixed learning rate of 0.01 across all 100 epochs, SGD does not benefit from Adam’s adaptive per-parameter step sizes, which appear to be particularly useful for the sparse gradient structure produced by L_1 regularisation on the interaction weights. The practical implication is that, for interaction detection on these benchmark functions optimizer choice is a meaningful source of performance variation comparable in magnitude to the effect of moderate noise.

Table 9: Mean AUPRC by optimizer, averaged across all other conditions.

Optimizer	Mean AUPRC	SD
Adam	0.513	0.208
SGD	0.429	0.171

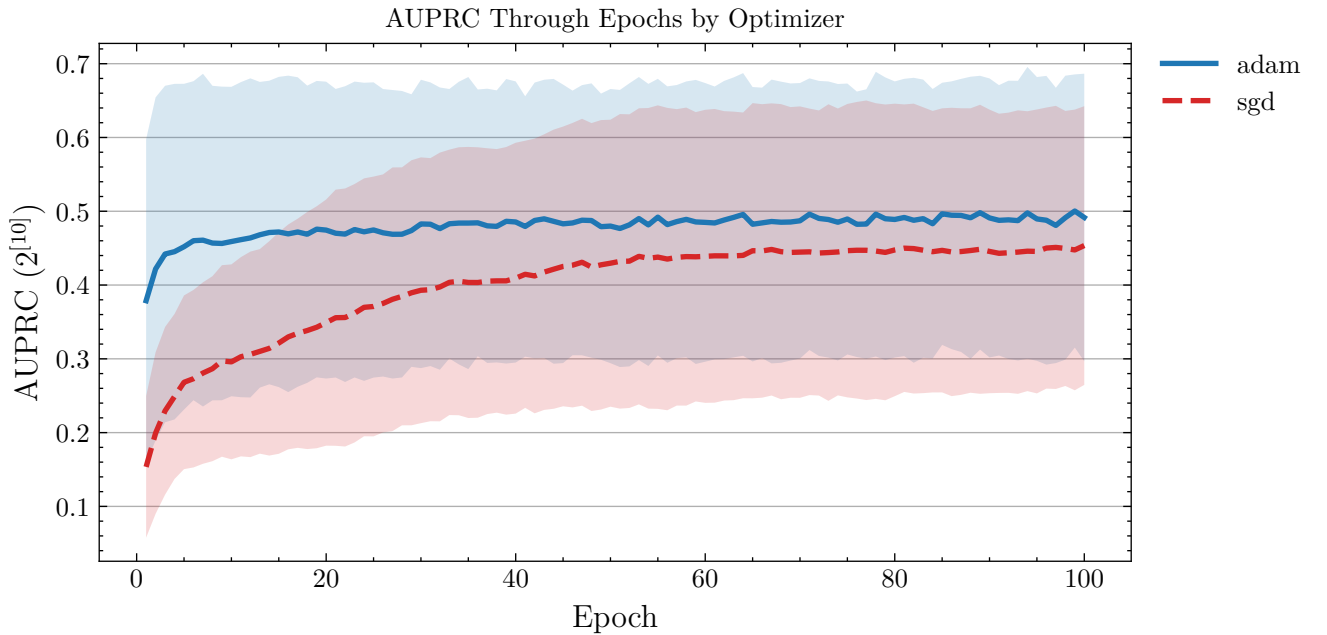


Figure 11: AUPRC trajectories over 100 epochs by optimizer, averaged across all functions and experimental conditions. Shaded regions denote ± 1 standard deviation across functions.

4.9 Architecture Comparison

Table 10 and Figure 12 show that neither architecture dominates consistently across functions. Of the ten benchmark functions, the small model achieves higher mean AUPRC on seven ($f_1, f_3, f_5, f_7, f_8, f_9, f_{10}$) while the big model leads on three (f_2, f_4, f_6). The margins vary considerably: for f_5 and f_8 the small model outperforms the big model by 0.133 and 0.141 points respectively, while for f_2, f_4 , and f_6 the big model’s advantages are modest (0.021, 0.030, and 0.039).

The consistent advantage of the small model on the higher-performing functions (f_5, f_8, f_{10}) is noteworthy. These are also the functions with the largest standard deviations in Table 5, indicating that they are the most sensitive to experimental conditions. The pattern is consistent with the noise analysis in Section 4.4: greater model capacity appears to increase susceptibility to label noise and regularisation, preventing the big model from reliably reaching the high AUPRC values that the small model achieves under favourable conditions. On the lower-performing functions (f_1, f_2, f_3, f_4), differences between architectures are much smaller, suggesting that on structurally difficult functions, capacity alone is not the limiting factor.

Taken together, the architecture comparison does not support a straightforward recommendation in favour of deeper networks for interaction detection. The big architecture offers no systematic advantage and incurs meaningful costs on functions where interaction recovery would otherwise be higher.

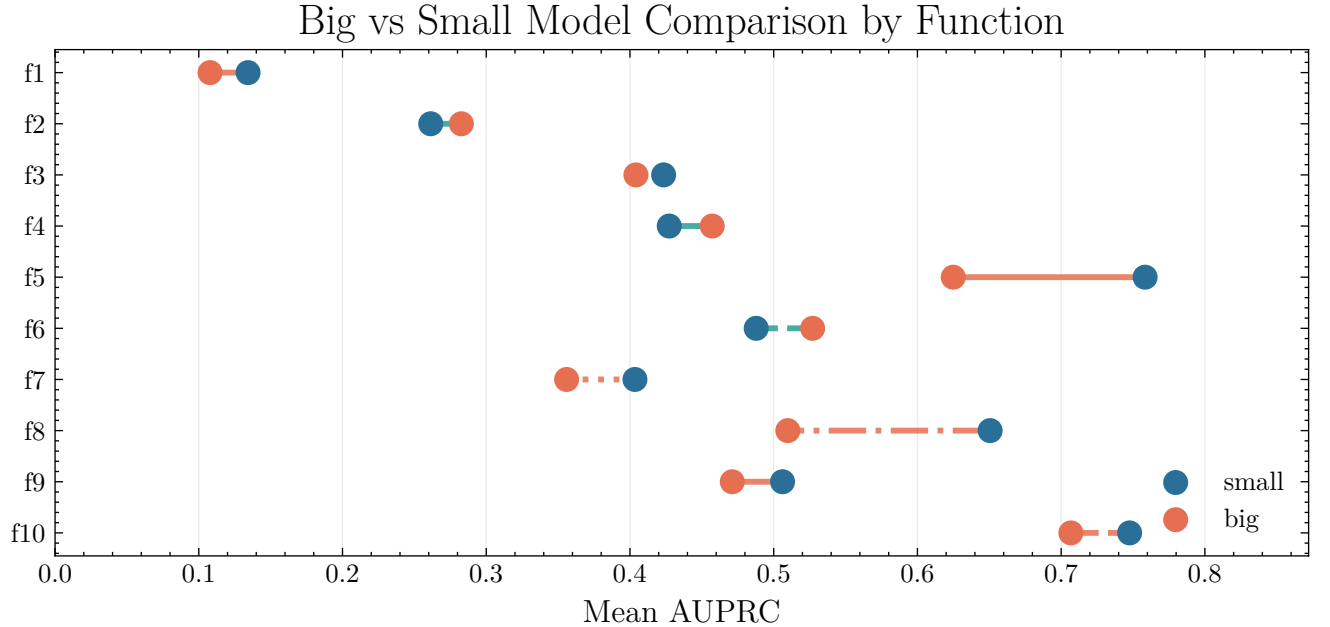


Figure 12: Mean AUPRC ($2^{[10]}$) at best epoch for the small and big architectures across all ten synthetic functions, averaged across all experimental conditions.

Table 10: Per-function AUPRC comparison between small and big architectures, averaged across all experimental conditions.

Function	Small	Big	Δ (Big – Small)	Winner
f_1	0.134	0.108	−0.026	Small
f_2	0.261	0.283	+0.021	Big
f_3	0.423	0.404	−0.019	Small
f_4	0.427	0.457	+0.030	Big
f_5	0.758	0.625	−0.133	Small
f_6	0.488	0.527	+0.039	Big
f_7	0.403	0.356	−0.048	Small
f_8	0.651	0.510	−0.141	Small
f_9	0.506	0.471	−0.035	Small
f_{10}	0.748	0.707	−0.041	Small

4.10 AUPRC and Validation Loss

Figure 13 plots mean AUPRC against mean validation loss at the best epoch for each function–architecture combination. The Spearman correlation is $\rho = -0.22$, $p < 0.001$, computed on all individual convergent runs ($n = 626$). The negative sign confirms that lower validation loss is associated with better interaction recovery, but the magnitude is modest: validation loss explains only a small fraction of the variance in AUPRC. The multi-seed replication (Section 4.11) recovers the same qualitative result: a weak negative correlation ($\rho = -0.140$) whose 95% bootstrap confidence interval $[-0.229, -0.056]$ excludes zero, confirming that the association is genuine in sign while remaining weak in magnitude.

The scatter makes this clear. Several function–architecture combinations achieve high AUPRC (above 0.65) at low validation loss, while others with similarly low validation loss remain near 0.25–0.40. In the mid-range of validation losses (10^{-1} to 10^0), the spread in AUPRC is almost as large as the full observable range. This suggests that a model can converge to a low training objective while encoding very different amounts of interaction structure in its weights, depending on the function, the regularisation applied, and the architecture.

The practical implication for interaction detection is that validation loss is a weak proxy for interaction recovery quality and cannot be used reliably as a selection criterion for the best checkpoint or the best experimental condition. Evaluation against the ground truth, as done throughout this study, is necessary to distinguish conditions that genuinely improve interaction recovery from those that merely reduce the training objective.

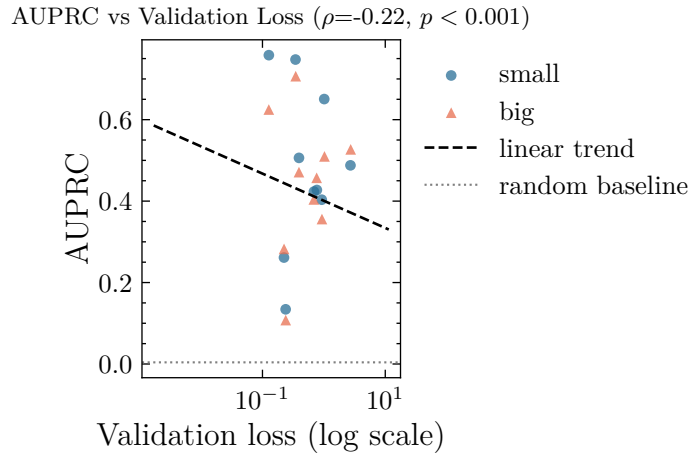


Figure 13: AUPRC at best epoch versus validation loss for each function and architecture. Each point represents the mean over all convergent runs for one function–architecture combination. The dashed line shows the linear trend fitted in log-space. Spearman $\rho = -0.22$, $p < 0.001$, computed on all individual convergent runs ($n = 626$), not on the aggregated points shown.

4.11 Multi-Seed Significance of Main Findings

All results reported so far derive from single-seed runs, which conflate genuine structural variance between functions with run-to-run randomness (Section 4.6). To test whether the main findings hold across multiple seeds, the core grid was replicated across 5 independent seeds, with the seed varied consistently across data generation, weight initialisation, and the train/validation split. This covers four regularisation conditions crossed with three noise levels ($\eta \in \{0, 0.5, 1.0\}$) on the small architecture with Adam, across all 10 functions, for 600 runs; 461 (76.8%) were retained after the same 3% convergence filter. Comparisons are paired by (seed, function) and tested with Wilcoxon signed-rank tests; the regularisation and noise family is Holm–Bonferroni corrected. Full results are in Table 11.

Table 11: Significance tests for main findings under multi-seed replication (5 seeds, small architecture, Adam; core grid). Comparisons are paired by (seed, function) and tested with Wilcoxon signed-rank tests, the noise effect uses a Friedman test. Claims A through D are jointly Holm–Bonferroni corrected; the Spearman correlation is reported uncorrected. $\Delta\tilde{m}$ denotes the median paired difference in AUPRC, r the rank-biserial effect size.

Claim	Comparison	n	Effect size	p_{adj}
A	Dropout vs. base	46	$\Delta\tilde{m} = -0.052, r = -0.509$	0.010
B	Dropout + WD vs. base	45	$\Delta\tilde{m} = -0.114, r = -0.767$	< 0.001
C	Weight decay vs. base	50	$\Delta\tilde{m} = +0.000, \text{CI } [-0.014, +0.000]$	0.424
D	Noise effect (base)	33	$\chi_F^2 = 29.29$	< 0.001
D	Noise acceleration	33	$\Delta\tilde{m}_{\text{accel}} = +0.142$	0.010
E	AUPRC vs. val. loss	461	$\rho = -0.140, \text{CI } [-0.229, -0.056]$	0.003 *

* Uncorrected; the Spearman test is excluded from the Holm–Bonferroni family (see Section 4.11).

The regularisation findings replicate. Dropout significantly reduces recovery relative to base (median $\Delta = -0.052, r = -0.509, p_{\text{adj}} = 0.010$), confirming the single-seed result that dropout suppresses the weight structure NID reads. Weight decay, by contrast, is statistically and practically negligible: the median difference from base is $+0.000$ with a 95% CI of $[-0.014, +0.000]$ and no significant effect ($p = 0.424$), supporting the interpretation that L_2 penalisation constrains weight magnitude without dismantling the relative path-strength ordering.

The combined dropout and weight-decay condition shows the largest reduction of all (median $\Delta = -0.114, r = -0.767, p_{\text{adj}} = < 0.001$). Note that the replication covers a reduced noise set 0, 0.5, 1.0; over these levels the combined condition is reliably worse than dropout alone, a gap driven primarily by the noise-free setting (Figure 9), whereas at the intermediate noise levels excluded from the replication the two conditions are closer. The single-seed near-equivalence therefore reflects averaging over those intermediate levels; once they are set aside, weight decay adds measurable degradation on top of dropout rather than being inert.

The noise findings also replicate. A Friedman test across the three noise levels in the base condition is significant ($\chi_F^2 = 29.29, p_{\text{adj}} = < 0.001$), and a targeted contrast confirms that the decline accelerates: the $0.5 \rightarrow 1.0$ drop exceeds the $0 \rightarrow 0.5$ drop (median acceleration $+0.142, p_{\text{adj}} = 0.010$). This supports describing the noise effect as an accelerating decline rather than a sharp threshold.

Finally, the weak association between validation loss and recovery persists under replication (Spearman $\rho = -0.140$, 95% bootstrap CI $[-0.229, -0.056]$). The interval excludes zero, confirming the direction, while the small magnitude reinforces that validation loss is an unreliable selection criterion for interaction recovery.

5 Discussion

The objective of this paper was to find how regularization techniques and label noise affect the trajectory and stability of interaction recovery in multi-layer perceptrons, evaluated against ground truth. The results have allowed several conclusions to be drawn, but also reveal a number of limitations that limit the scope of the findings.

Regularisation does not uniformly improve interaction recovery

A consistent finding throughout all experiments is that regularization does not improve interaction recovery in comparison to an unregularized baseline. Dropout lowers mean AUPRC compared to the baseline by 22.9% and dropout in combination with weight decay has a very similar reduction of 23.7%.

This is consistent with Lengerich et al. (2021), who observe that dropout causes the model to break higher-order interactions, thus lowering the ability of the model to learn these interactions. In the context of interaction detection, this means that the very mechanism intended to prevent overfitting actively suppresses the weight structure that NID relies on to identify interactions.

Weight decay presents a different picture. Its reduction relative to base is modest (-7.1% averaged across noise levels), and the heatmap in Figure 9 shows that weight decay tracks base closely at low to moderate noise, even slightly outperforming it at $\eta = 0.5$ (0.47 vs 0.45). Only at the highest noise level does it fall below base (0.32 vs 0.39), consistent with the lower learnable signal there. This pattern

suggests that L_2 penalisation constrains weight magnitude without dismantling the relative strength differences between interacting and non-interacting paths, which are what NID ultimately reads.

In the single-seed results, dropout alone and dropout combined with weight decay produce very close mean AUPRC (0.404 and 0.400), which initially suggested that dropout is the dominant factor. This near-equivalence, however seems to disappear with multi-seed significance testing (Section 4.11). Under paired testing across seeds on the reduced noise set ($\eta \in \{0, 0.5, 1.0\}$), the combined condition produces the largest reduction of all (median $\Delta = -0.114$, $r = -0.767$), exceeding the effect of dropout alone (median $\Delta = -0.052$); the combined condition is worse or equal to dropout alone at each of these three noise levels rather than only in aggregate. This indicates that weight decay is not inert once dropout is present: the two regularisers compound, with weight decay adding further degradation on top of dropout, most clearly at the lowest and highest noise levels ($\eta = 0$ and $\eta = 1$). Dropout remains the principal disruptor in both analyses, but the single-seed near-equivalence appears to understate the cost of combining the two.

Output noise lowers the recovery ceiling without changing early dynamics

Output noise degrades interaction recovery monotonically, but the trajectories in Figure 7 suggest that this degradation is better characterised as a ceiling effect than as a disruption of the recovery process itself. All noise levels share the same rapid early rise in the first 10–20 epochs; the trajectories then diverge to different plateaus that they maintain stably for the remainder of training.

These results are in agreement with Arpit et al. (2017), who found that models trained on noisy label data tend to learn real patterns at first before learning noise. If this order is the same, interaction structures are encoded in the weights in the initial phase of training before noise starts to be encoded in later training phases.

Recovery appears relatively robust to low label noise but degrades more readily once noise exceeds a moderate level: the pooled decline is gradual through $\eta = 0.2$ and then steepens markedly, with the interval between $\eta = 0.2$ and $\eta = 0.5$ accounting for the largest single increase in degradation. Caution should be exercised before reading this as a sharp threshold or regime change, given the substantial between-function variance, but the pattern is at least consistent with noise variance progressively overpowering the interaction signal that NID relies on. The two architectures reinforce a familiar capacity–robustness trade-off: the larger model achieves the strongest recovery in the noise-free setting yet falls below the smaller model under any noise, and the gap widens as noise grows. This suggests that added capacity aids interaction recovery only when the signal is clean, and becomes a liability as label noise increases the possibility to overfit on the noisy signal.

Validation loss is a weak proxy for interaction recovery

One of the more relevant findings is that validation loss is a poor predictor of interaction recovery quality. The Spearman correlation between the two quantities, computed across all 626 convergent runs, is $\rho = -0.22$, $p < 0.001$. The fact that the correlation is negative confirms that lower validation loss is generally associated with a higher AUPRC. However, the magnitude of this correlation is weak, and the scatter plot in Figure 13 shows that the models achieve AUPRCs spanning almost the full observable range.

This was further confirmed in section 4.11 which found a weak negative correlation under multi-seed replication ($\rho = -0.140$, $CI_{95\%} = [-0.229, -0.056]$). The interval excludes zero, establishing that the negative association is genuine rather than an artefact of a single seed, while its small magnitude reinforces that validation loss tracks interaction recovery only weakly.

This finding is something to keep in mind when using NID, or any similar weight-based method. Model chosen for its validation loss will not reliably be the model with the highest interaction recovery. Evaluation against ground truth, where available, is therefore a necessary complement to standard loss-based model selection.

Smaller models recover interactions more reliably under adverse conditions

The architecture comparison reveals no systematic advantage for the deeper, wider big model. The small model achieves higher mean AUPRC on seven of ten benchmark functions, and the largest margins are precisely on the functions where recovery is otherwise high (f_5 , f_8 , f_{10}). This is consistent with the noise analysis: the big model’s greater capacity appears to increase its susceptibility to label noise and

regularisation, preventing it from reliably reaching the recovery levels that the small model attains with the right conditions.

More complicated functions with lower average AUPRC on both big and small architectures have very close interaction recovery. This entails that model capacity has little effect on the ability of the model to capture interactions on these more complex functions. We can assume that, in these experimental conditions, interaction detection signals are weak enough to cause larger models to capture spurious patterns and noise instead of finding ground truth interactions.

Limitations

These findings are subject to several limitations. First, outside the main results, some findings are based on single-seed runs, so the variability across functions reported throughout Section 4 merges genuine structural differences between functions with run-specific randomness. To address this directly, a multi-seed replication of the core grid was conducted (Section 4.11), confirming the dropout, weight-decay, and noise findings under paired significance testing across 5 seeds. This replication was, however, restricted to the small architecture with Adam and the reduced noise set $\eta \in \{0, 0.5, 1.0\}$; the optimizer and architecture comparisons were not re-run across seeds, so claims about those two factors remain single-seed. A full multi-seed replication of the entire experimental design would be needed to extend the stability guarantees to every result.

Second, the exclusive use of benchmark synthetic functions specifically designed to test interactions may give results that are not reproducible using real-world data. Results should not be assumed to generalise directly to applied settings.

Third, NID is treated as the only interaction detection method. The sensitivity of the findings to this choice is not assessed. Alternative detection methods, such as those based on Shapley interaction indices or neural additive models, may respond differently to the same regularization and noise conditions.

Fourth, the convergence filter described in Section 3.6 excludes runs where validation loss did not decrease by at least 3% across training. Even though this threshold is conservative, it may create a selection bias for models with more stable training. This might cause the results to underestimate the failure rate of heavily regularized or high-noise experiments.

Fifth, when classifying detected interactions ($\mathcal{D}$), supersets of ground truth interactions ($\mathcal{G}$) are classified as false positives. This is the strictest interpretation of exact interaction recovery. It penalizes cases when NID recovers the correct feature set embedded within a larger interaction. This may lead to underestimating partial interaction recovery in higher-order interaction functions.

Finally, the noise manipulation in this study adds Gaussian noise to the output variable y , not to the input features x . This is label or output noise, and its relationship to regularisation is not the same as for input noise. In particular, the formal equivalence between input Gaussian noise and L_2 regularisation established by Goodfellow et al. (2016) does not apply here. The effects of output noise on interaction recovery trajectories documented in this study should therefore be understood as distinct from the dynamics that would arise under input perturbation.

Several directions for future research follow directly from these limitations. Replicating the experiment with multiple random seeds would allow genuine function-level stability to be separated from run-to-run randomness, and would allow statistical significance testing across all experiments. Evaluating these experiments on real-world datasets with partially known interaction structures would further test the generality of the findings.

Comparing NID to other interaction detection instruments, such as Shapley or H-statistics, under the same noise and regularization conditions would establish whether the observed results are specific to weight-based methods or are a concrete limitation of recovery under label noise. Finally, exploring input noise as a distinct experimental factor, by perturbing features rather than labels, would allow a direct empirical comparison between the two noise regimes rather than relying on the theoretical equivalence that motivates combining them.

6 Conclusion

This paper studies how regularization techniques and label noise affect trajectory and stability of interaction recovery in MLPs, using NID against ground truth.

The main finding is that regularisation does not improve interaction recovery. Averaged across all noise levels, functions, and architectures, dropout reduces mean AUPRC from the unregularised baseline by 22.9%, whereas weight decay is much more benign, falling only to 7.1%. The combination of both weight decay and dropout changes recovery negligibly in the single seed analysis (0.404 to 0.400). When multiple-seed analysis is applied, the combination of weight decay and dropout appears to have a more cumulative effect (weight decay alone produces a median change of +0.000, whereas combining it with dropout yields -0.114) especially at noise level 0 and 1. These results suggest that standard regularisation practice transfers poorly to the interaction detection setting: the mechanisms that prevent overfitting in prediction tasks can simultaneously prevent the network from encoding the weight structure that NID relies on.

Output noise lowers the attainable recovery plateau without fundamentally altering the early trajectory shape, consistent with the view that interaction structure is encoded during the first 10–20 epochs of training before noise fitting begins to compete for representational capacity. Pooled recovery declines gradually through $\eta = 0.2$ and then more steeply, with the interval between $\eta = 0.2$ and $\eta = 0.5$ marking the largest single drop; recovery continues to fall at a comparable rate to $\eta = 1$, so the pattern is better described as an accelerating decline than as a sharp threshold, particularly given the substantial between-function variance. The two architectures degrade similarly, though the larger model, despite a marginal advantage in the noise-free setting falls below the small model at every positive noise level and is lowest at $\eta = 1$, indicating that its extra capacity aids recovery only when the signal is clean.

Taken together, these findings suggest that practitioners seeking to use weight-based interaction detection should exercise caution when applying standard regularisation schemes, prefer smaller over larger architectures when interpretability is the primary goal, and evaluate interaction recovery directly against available ground truth rather than relying on validation loss as a proxy. Future work replicated across multiple seeds, extended to real-world datasets, and comparing alternative detection instruments would strengthen these conclusions and clarify their scope.

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A Run-Level Convergence Exclusions

Figure 14 shows the full run-level breakdown of convergence exclusions across all ten synthetic functions and all experimental conditions. Each row corresponds to one experiment (identified by optimiser and regularisation condition, grouped by noise level), and each column corresponds to one synthetic function. A filled circle indicates that the small model run was excluded; a filled square indicates that the big model run was excluded. Empty cells indicate that the run passed the convergence criterion and was retained.

The plot confirms that exclusions are not systematically concentrated on particular functions, suggesting that non-convergence reflects the difficulty of the experimental condition rather than properties of specific synthetic functions.

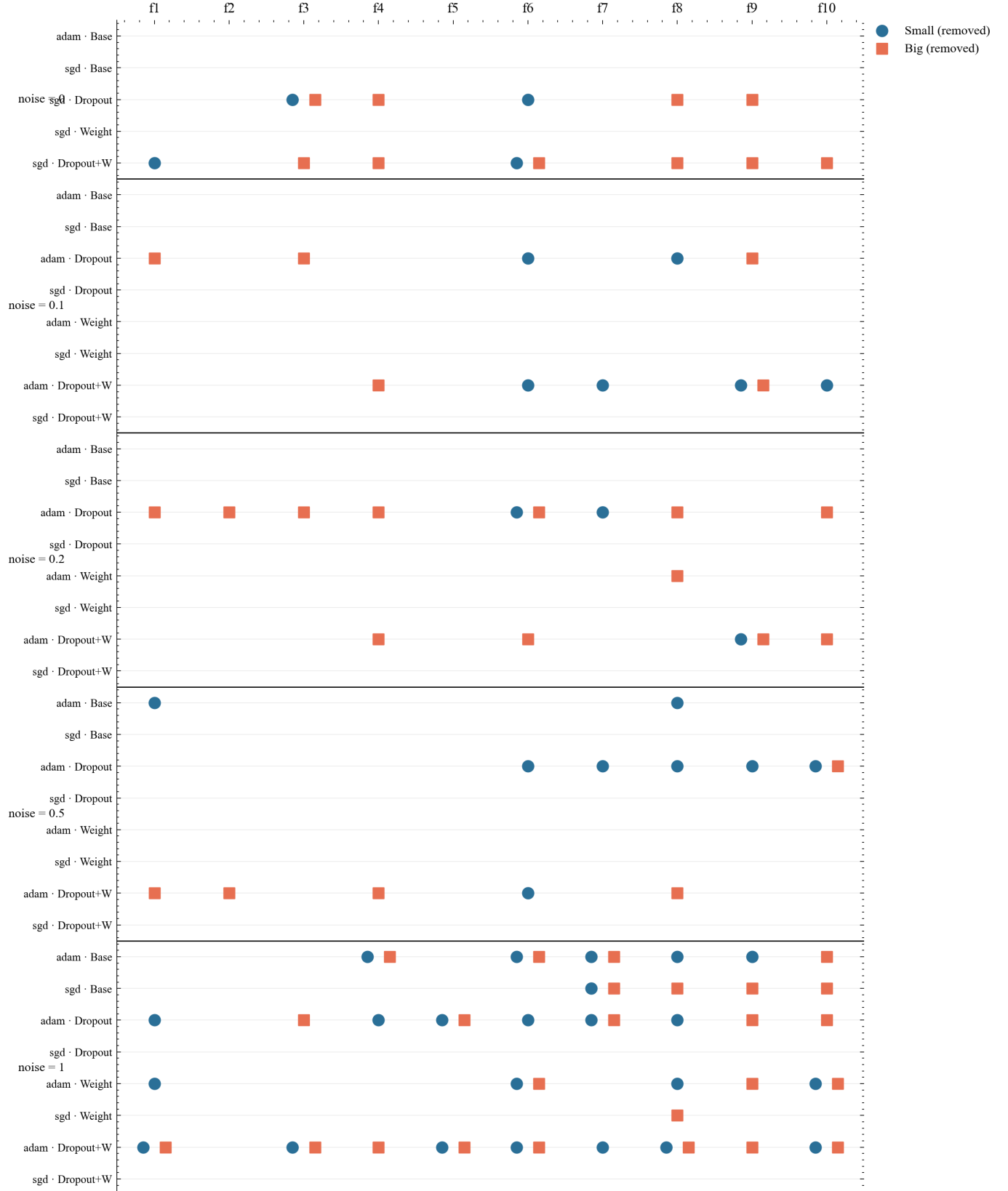


Figure 14: Run-level convergence exclusions across all experimental conditions and synthetic functions. Rows are grouped by noise level and sorted by regularisation condition and optimiser within each group. A filled circle (●) indicates the small model run was excluded; a filled square (■) indicates the big model run was excluded. Empty cells indicate the run was retained. Exclusions are distributed across functions without strong systematic concentration, suggesting non-convergence is driven by experimental conditions rather than function-specific properties.

B Full Interaction Recovery Trajectories

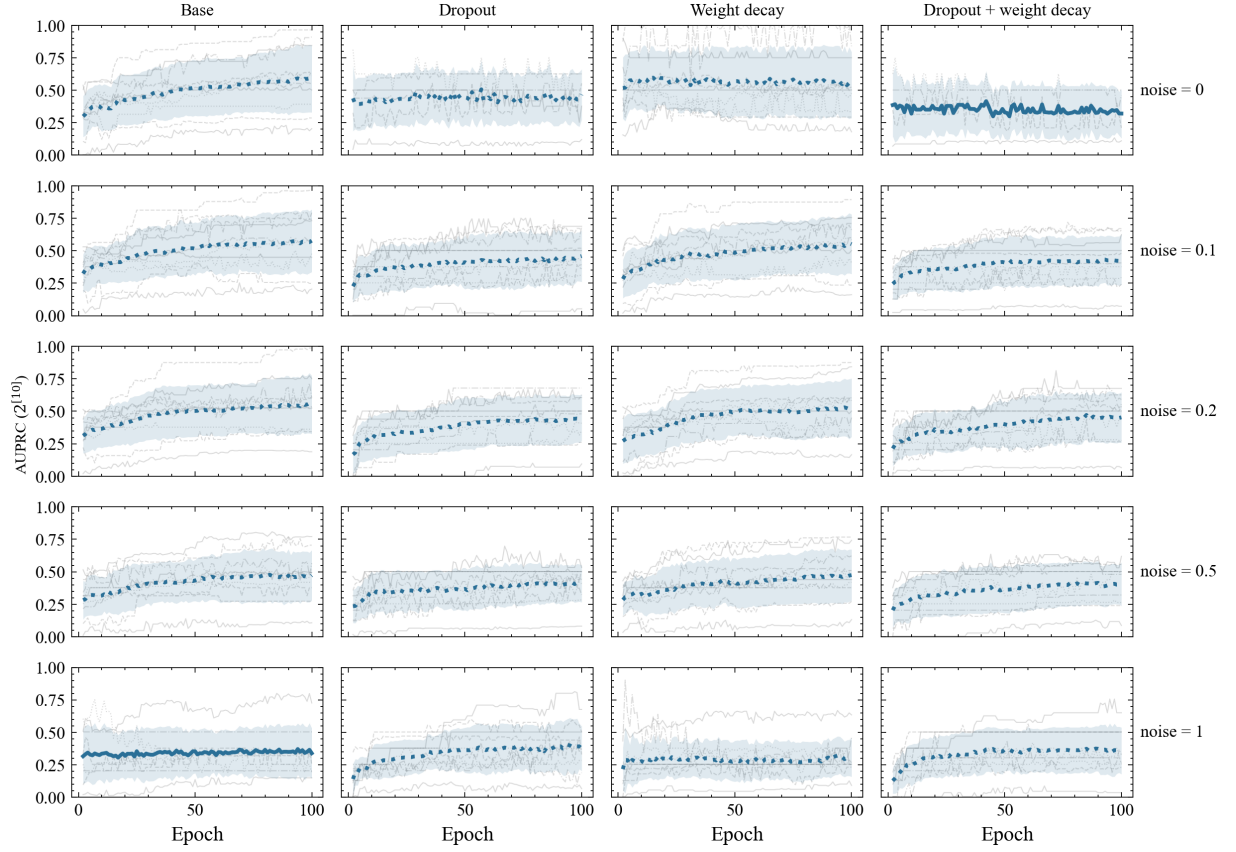


Figure 15: AUPRC (2^{10}) recovery trajectories over 100 epochs for all combinations of noise level (rows) and regularisation condition (columns). Grey lines show individual function trajectories; the bold line shows the mean across functions with ± 1 standard deviation shaded.

C Regularisation Dose–Response Detail

Table 12 reports the per-condition best-epoch AUPRC and percentage change relative to the zero-strength anchor for the dropout and weight-decay sweeps described in Section 4.3.

Table 12: Best-epoch AUPRC dose-response for single-seed probe sweep (functions f1, f7, f8; small architecture; $\eta = 0$; Adam). Mean % change is per-function normalised before averaging across functions. Dropout shape: *flat then collapse*. L2 weight-decay shape: *mixed response*.

Constant	Mean AUPRC	SD	Mean % change
<i>Dropout (anchor: $p = 0$)</i>			
$p = 0$	0.409	0.235	+0.0
$p = 0.1$	0.376	0.315	-14.7
$p = 0.2$	0.401	0.227	+9.6
$p = 0.3$	0.360	0.266	-2.1
$p = 0.5$	0.240	0.168	-39.6
<i>L2 weight decay (anchor: $\lambda = 0$)</i>			
$\lambda = 0$ (anchor)	0.523	0.414	+0.0
$\lambda = 10^{-5}$	0.579	0.366	+19.8
$\lambda = 10^{-4}$	0.515	0.424	-0.3
$\lambda = 10^{-3}$	0.503	0.430	-6.3
$\lambda = 10^{-2}$	0.331	0.149	+1.5

D Official Statement of original thesis

Appendix:

Official statement of original thesis

By signing this statement, I hereby acknowledge the submitted thesis (hereafter mentioned as "product"), titled:

..... How do regularization techniques and output noise influence the trajectory and stability of interaction recovery in MLPs, as measured against ground truth?

to be produced independently by me, without external help.

Wherever I paraphrase or cite literally, a reference to the original source (journal, book, report, internet, etc.) is given.

By signing this statement, I explicitly declare that I am aware of the fraud sanctions as stated in the Education and Examination Regulations (EERs) of the SBE.

Place: Maastricht

Date: June 5th 2026

First and last name: Edouard Dewaerheijd

Study programme: Business Analytics

EBT Code: EBT0017

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